Let us define:

$$\omega = e^{\frac{i\pi}{5}}, \qquad \phi = \frac{1+\sqrt{5}}{2}$$

We want to show that:

$$L = \frac{\left|\omega^2 a\left(1+\phi+\omega^3\phi(1+\phi)\right) - \sqrt{\phi}b\left(1+\phi-\omega^3(1+\phi)\right)\right|^2}{\phi(1+\phi)^4} + \frac{\left|\omega^2\phi a\left(1+\phi+\omega^3\phi(1+\phi)\right) - \sqrt{\phi^3}b\left(1+\phi-\omega^3(1+\phi)\right)\right|^2}{\phi^4(1+\phi)^4}$$

Is equal to

$$R = \frac{\left|a+\sqrt{\phi}b\right|^2}{(1+\phi)^2} + \frac{\left|\sqrt{\phi}a+\phi b\right|^2}{(1+\phi)^2}$$

Simplifying L using $1+\phi = \phi^2$:

$$L = \frac{\left|\omega^2 a(\phi^2+\omega^3\phi^3) - \sqrt{\phi}b(\phi^2-\omega^3\phi^2)\right|^2}{\phi^9} + \frac{\left|\omega^2\phi a(\phi^2+\omega^3\phi^3) - \sqrt{\phi^3}b(\phi^2-\omega^3\phi^2)\right|^2}{\phi^{12}}$$

$$= \frac{|\phi|^4\left|\omega^2 a(1+\omega^3\phi) - \sqrt{\phi}b(1-\omega^3)\right|^2}{\phi^9} + \frac{|\phi|^6\left|\omega^2 a(1+\omega^3\phi) - \sqrt{\phi}b(1-\omega^3)\right|^2}{\phi^{12}}$$

$$= \frac{\left|\omega^2 a(1+\omega^3\phi) - \sqrt{\phi}b(1-\omega^3)\right|^2}{\phi^5} + \frac{\left|\omega^2 a(1+\omega^3\phi) - \sqrt{\phi}b(1-\omega^3)\right|^2}{\phi^6}$$

$$= \frac{\left|\omega^2 a(1+\omega^3\phi) - \sqrt{\phi}b(1-\omega^3)\right|^2}{\phi^5}\underbrace{\left(1+\frac{1}{\phi}\right)}_{=\phi}$$

$$= \frac{\left|\omega^2 a(1+\omega^3\phi) - \sqrt{\phi}b(1-\omega^3)\right|^2}{\phi^4}$$

Using $2\cos\left(\frac{\pi}{5}\right) = \omega + \omega^{-1} = \phi$:

$$L = \frac{\left|\omega^2 a\left(1+\omega^3(\omega+\omega^{-1})\right) - \sqrt{\phi}b(1-\omega^3)\right|^2}{\phi^4}$$

$$= \frac{\left|\omega^2 a(1+\omega^4+\omega^2) - \sqrt{\phi}b(1-\omega^3)\right|^2}{\phi^4}$$

Using the fact that ω is the fifth root of unity and therefore is a root to the 5th cyclotomic polynomial, $\omega^4 - \omega^3 + \omega^2 - \omega + 1 = 0 \Rightarrow 1+\omega^4+\omega^2 = \omega^3+\omega$:

$$L = \frac{\left|\omega^2 a(\omega+\omega^3) - \sqrt{\phi}b(1-\omega^3)\right|^2}{\phi^4}$$

$$= \frac{\left|a(\omega^3-1) - \sqrt{\phi}b(1-\omega^3)\right|^2}{\phi^4}$$

$$= |\omega^3-1|^2\left(\frac{\left|a+\sqrt{\phi}b\right|^2}{\phi^4}\right)$$

Lastly, note that

$$|\omega^3 - 1| = \left|\omega^{\frac{3}{2}} - \omega^{-\frac{3}{2}}\right| = \left|e^{\frac{3\pi i}{10}} - e^{-\frac{3\pi i}{10}}\right| = 2\sin\left(\frac{3\pi}{10}\right) = 2\cos\left(\frac{2\pi}{10}\right) = 2\cos\left(\frac{\pi}{5}\right) = \phi$$

Thus:

$$L = \frac{\phi^2}{\phi^4}\left|a + \sqrt{\phi}b\right|^2 = \frac{1}{\phi^2}\left|a + \sqrt{\phi}b\right|^2 = \left|\frac{a + \sqrt{\phi}b}{\phi}\right|^2$$

Simplifying R:

$$\begin{aligned} R &= \frac{\left|a + \sqrt{\phi}b\right|^2}{(1+\phi)^2} + \frac{\left|\sqrt{\phi}a + \phi b\right|^2}{(1+\phi)^2} \\ &= \frac{\left|a + \sqrt{\phi}b\right|^2}{\phi^4} + \frac{\phi\left|a + \sqrt{\phi}b\right|^2}{\phi^4} \\ &= \frac{\left|a + \sqrt{\phi}b\right|^2}{\phi^4}(1+\phi) \\ &= \frac{\left|a + \sqrt{\phi}b\right|^2}{\phi^2} \\ &= \left|\frac{a + \sqrt{\phi}b}{\phi}\right|^2 \end{aligned}$$